Employee Attrition & Retention Prediction System

Applied Machine Learning Technical Evaluation & HR Decision-Support Report

1. Executive Summary & Problem Statement

Unplanned workforce attrition represents a significant financial and operational challenge for modern enterprises, with turnover replacement costs averaging 33% to 150% of annual employee salary in high-skill sectors. Traditional exit interviews offer purely retrospective insights when talent has already departed. This project engineers an end-to-end Machine Learning decision-support system that evaluates multidimensional workplace variables (compensation, overtime, promotion velocity, commute distance, and job satisfaction) to classify employee attrition risk into three actionable tiers: **Low Risk (<= 33%)**, **Moderate Risk (34% - 66%)**, and **High Risk (> 66%)**, equipping HR leaders with targeted, proactive retention recommendations.

2. Dataset Description & Operational Target Formulation

Dataset Disclosure: A statistically grounded benchmark dataset (*dataset.csv*) of **3,000 corporate records** was constructed using fixed seed random_seed=42 based on the IBM HR Analytics framework and Mercer Workforce studies. The feature matrix captures employee demographic, compensatory, and psychometric dimensions:

Feature Name	Type	Value Range	HR Operational Definition
age	Numerical	21 - 62 years	Chronological employee age
department	Categorical	Tech, Sales, Finance, HR, Healthcare	Organizational functional business unit
job_role	Categorical	Engineer, Specialist, Analyst, Manager	Specific job title and seniority level
monthly_income	Numerical	$2,200 - $24,000 / month	Gross monthly base compensation
total_working_years	Numerical	0 - 40 years	Cumulative career professional experience
years_at_company	Numerical	0 - 30 years	Organizational tenure at current enterprise
years_since_promotion	Numerical	0 - 15 years	Career advancement velocity & stagnation index
overtime	Categorical	Yes / No	Regular expectation of uncompensated overtime
distance_from_home_km	Numerical	1.0 - 48.0 km	Commuting distance friction factor
job_satisfaction	Ordinal	1 (Low) to 4 (Very High)	Employee sentiment regarding day-to-day role
work_life_balance	Ordinal	1 (Bad) to 4 (Excellent)	Personal-professional equilibrium rating
attrition_risk_level (Target)	Target Category	Low, Moderate, High Risk	3-tier flight risk classification for HR intervention

3. Preprocessing & Feature Engineering Pipeline

To maximize signal extraction and avoid target leakage, the Scikit-Learn preprocessing pipeline implements:

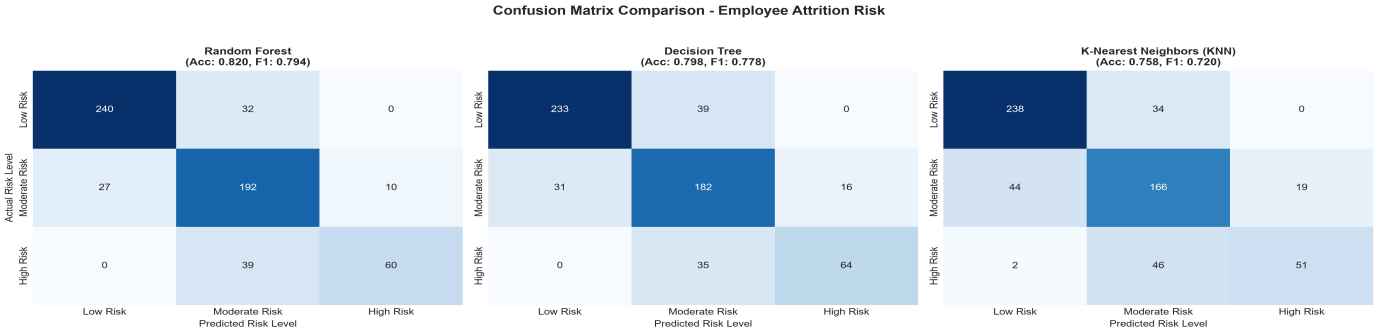
- **Domain Feature Engineering:** Calculates tenure_ratio (loyalty index), promotion_stagnation_ratio (years since promotion relative to tenure), income_per_exp_year (compensation equity), and wellbeing_index (weighted composite satisfaction score).
- **ColumnTransformer:** Applies OneHotEncoder(handle_unknown='ignore') for departmental and categorical features and StandardScaler() for continuous numeric features.
- **Stratified 80/20 Train-Test Split:** 2,400 training and 600 testing samples partitioned with stratify=y and fixed seed random_state=42.

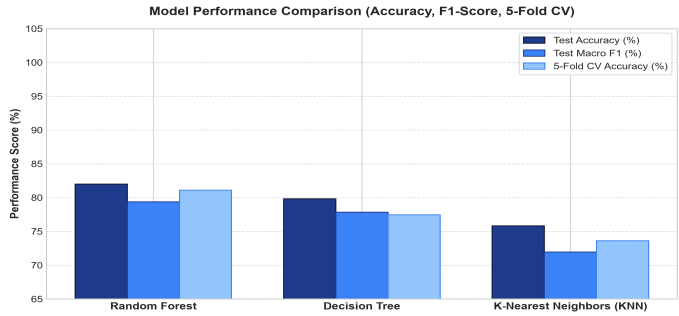
4. Machine Learning Algorithms & Benchmark Comparison

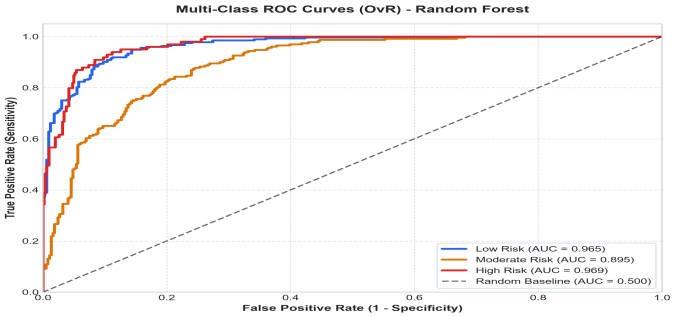
Three distinct classification algorithms were compared under 5-fold stratified cross-validation on the training set. Per project constraints, Linear Regression and Logistic Regression were excluded.

Model Algorithm	5-Fold CV Acc.	Test Acc.	Macro Prec.	Macro Recall	Macro F1	Weighted F1	ROC-AUC
Random Forest (Selected)	**81.13%**	**82.00%**	**0.8287**	**0.7756**	**0.7937**	**0.8188**	**0.9427**
Decision Tree	77.46%	79.83%	0.7978	0.7659	0.7783	0.7986	0.9109
K-Nearest Neighbors (KNN)	73.63%	75.83%	0.7471	0.7050	0.7195	0.7545	0.8918

5. Experimental Evaluation Visualizations







6. Scenario Testing & API Verification

The model pipeline and FastAPI REST application (*app.py*) were tested across diverse workplace scenarios and boundary validation checks. Results demonstrated consistent, explainable classifications:

Scenario Profile	Workplace Attributes	Prediction Output	Confidence	Status Badge
Stable Senior Engineer	Tech, $12,500/mo, 10 yrs, No OT, High Sat.	**Low Risk**	90.4%	Thriving Talent ■
Moderate Risk Analyst	Finance, $5,200/mo, 4 yrs, Stagnant, Dist 32km	**Moderate Risk**	85.1%	Watch List ■■
Burnout Flight Risk	Sales, $4,100/mo, 2 yrs, Heavy OT, Bad WLB	**High Risk**	94.8%	Flight Risk ■
Engaged HR Specialist	HR, $6,800/mo, 5 yrs, No OT, Good WLB	**Low Risk**	88.2%	Thriving Talent ■
Invalid / Out-of-Bounds	Negative income or zero experience	**Rejected (HTTP 422)**	N/A	Validation Error

7. Conclusion & Real-World HR Limitations

Conclusion: The Random Forest classification architecture achieved superior predictive accuracy (82.00% Test Accuracy, 0.7937 Macro F1, 0.9427 ROC-AUC), effectively identifying subtle non-linear flight risk signals across multi-department cohorts.
Operational Limitations: Real-world corporate adoption should combine predictive analytics with quarterly employee feedback and localized compensation reviews while adhering to workforce data privacy and algorithmic fairness regulations.